

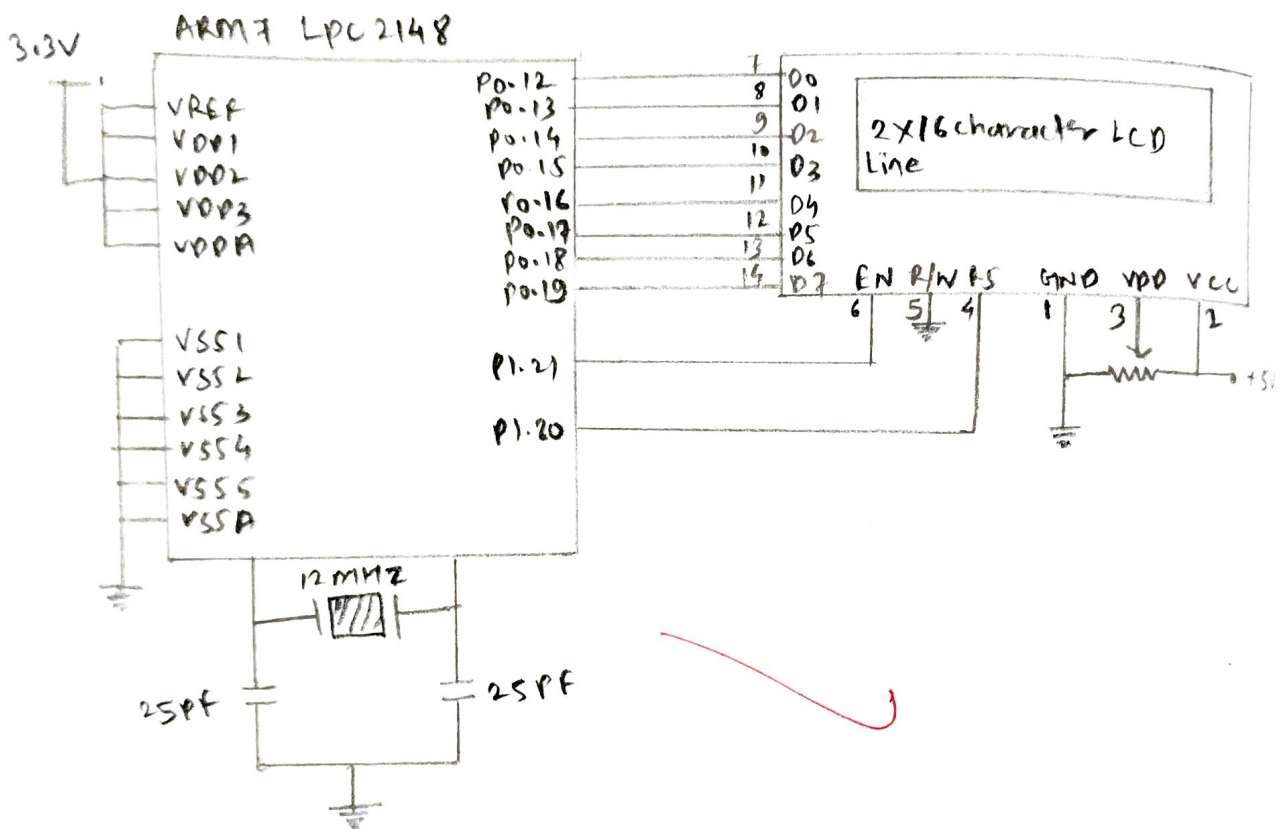


Fig. Interfacing diagram of LCD (2 Line x16 character) with LPC 2148 (ARM7).

The enable pin is used by the LCD to latch information. When data is supplied to data pins, a high to low pulse must be applied to the enable pin to latch the data present at the data pins. This pulse is called a "clock" or "enable" pulse.

1.1.5 Data bus (D0 - D7):

The 8-bit data pins, D0-D7 are used to send the information to the LCD's internal registers. To display the numbers and characters, the data pins are used while making RS=1. There are also instruction commands that can be sent to the display or blink the cursor.

We also use RS = 0 to check the busy flag bit to see if the LCD is ready to receive new information. The busy flag is D7 and can be read when R/W = 1. When the LCD is taking care of internal operations and will not accept any new information, it is ready to receive new information.

1.2 Pin Assignment of 16x2 LCD

Pin number	Symbol	Function
1	Vss	Power supply (GND)
2	Vcc	Power supply (+5V)
3	Vee	Contrast adjust
4	RS	0 = Instruction input, 1 = Data input
5	R/W	0 = Write to LCD memory, 1 = Read from LCD memory
6	E	Enable signal
7	DB0	Data bus line 0 (LSB)
8	DB1	Data bus line 1
9	DB2	Data bus line 2
10	DB3	Data bus line 3
11	DB4	Data bus line 4
12	DB5	Data bus line 5
13	DB6	Data bus line 6
14	DB7	Data bus line 7 (MSB)
15	VB+	Backlight Supply

2. Keyboard interfacing:

At the lowest level, keyboards are organized into rows and columns. When a key is pressed, the microcontroller accesses both rows and columns through ports. When the key is released, the contact is broken; otherwise, there is no connection between rows and columns. The microcontroller takes care of the keyboard debounce.

Interfacing Diagram: Students have to draw interfacing diagram.

Flowchart/ Algorithm : Students have to draw algorithm.

Department of E&TC, Sinhgad College of Engineering

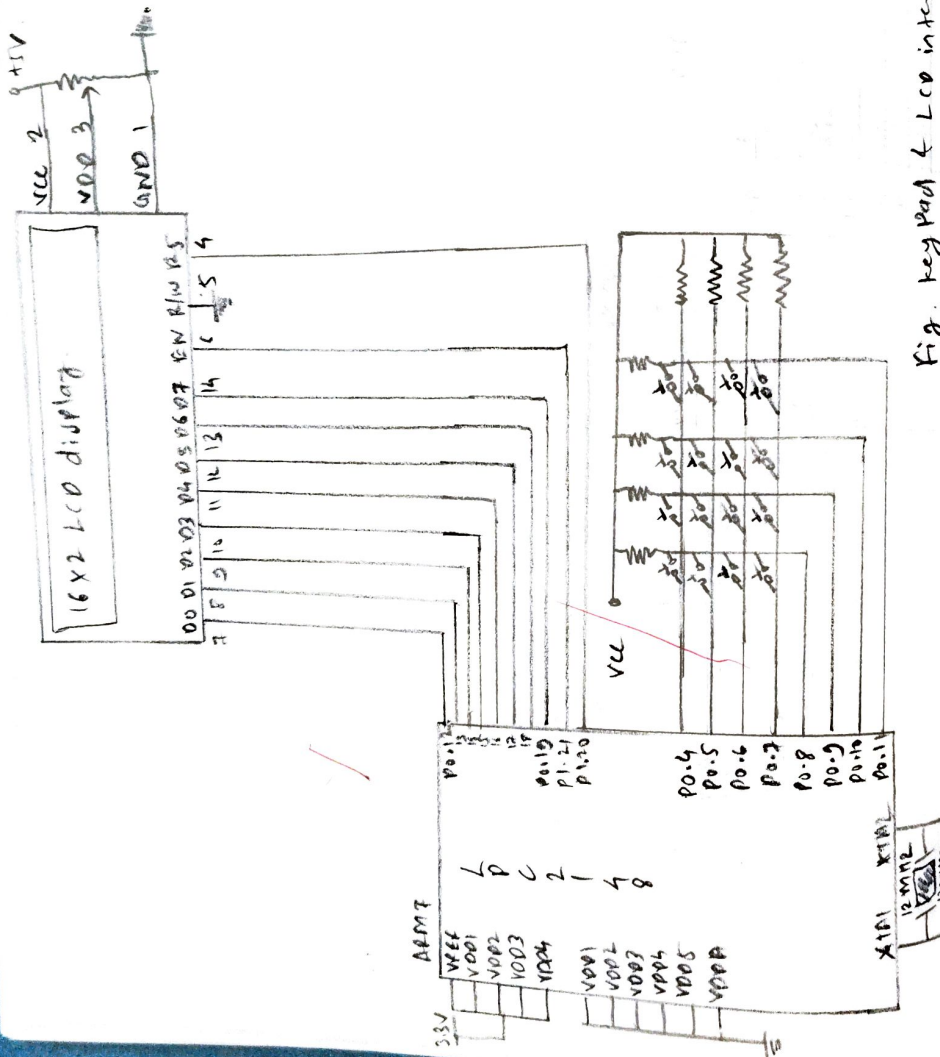


Fig. Key pad & LCD interfacing with ARM7, LPC2148.

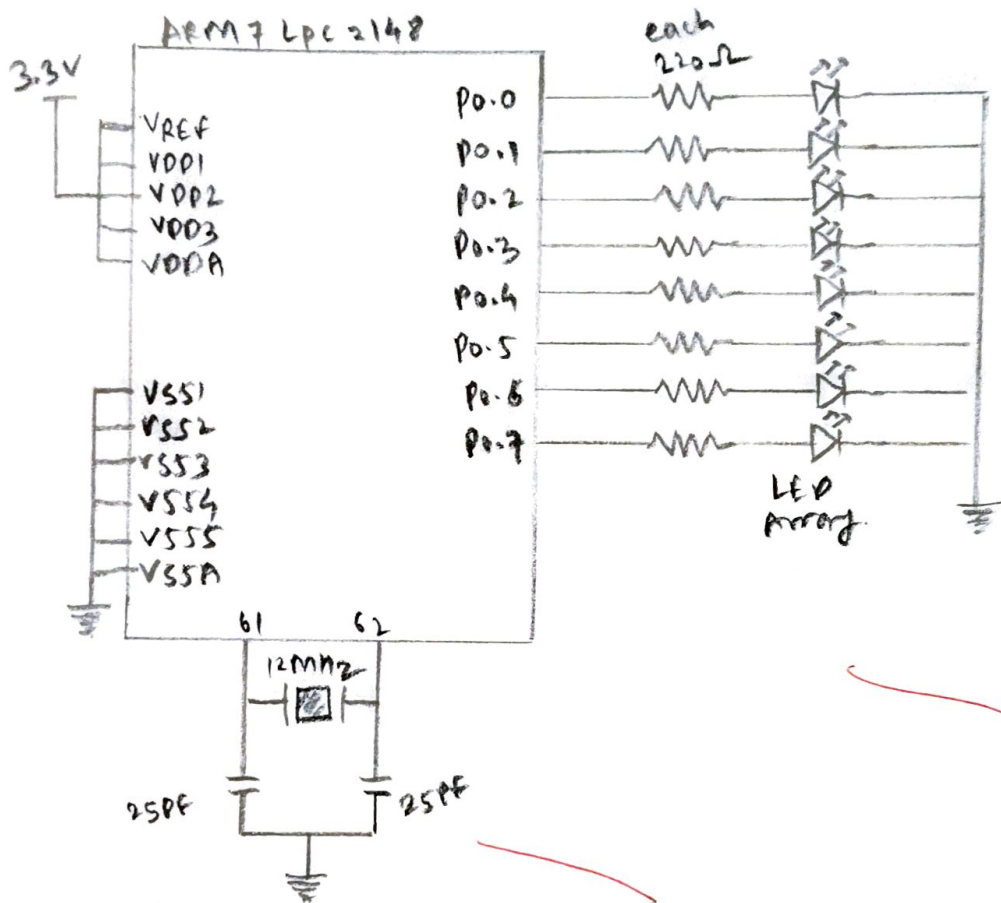


Fig Interfacing diagram of LED'S to (ARM7) LPC 2148.

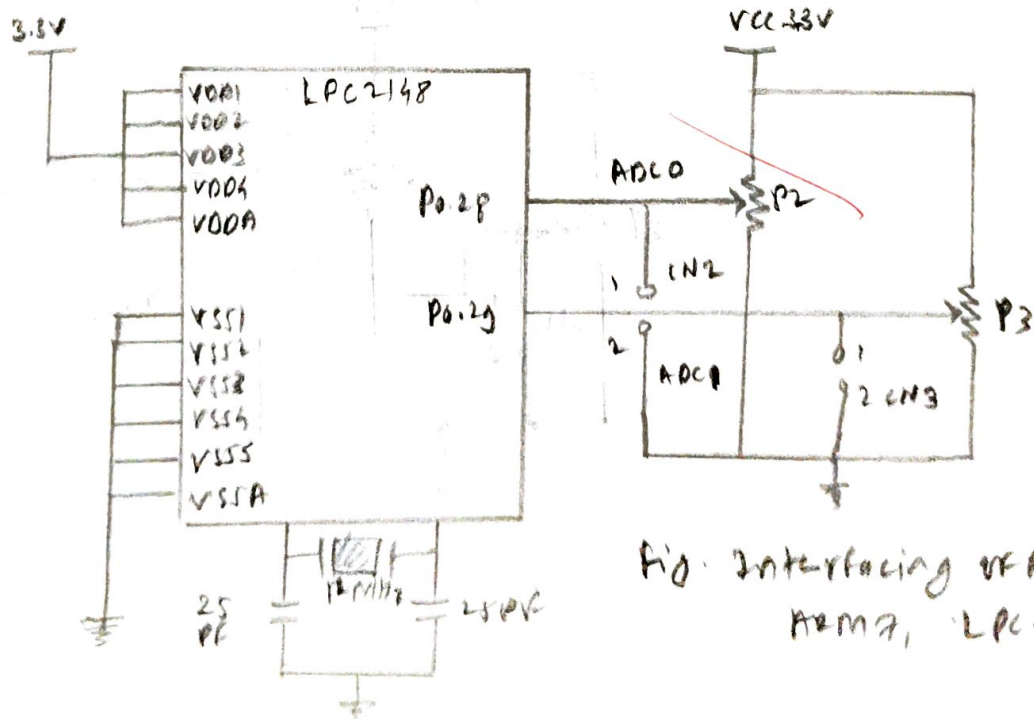
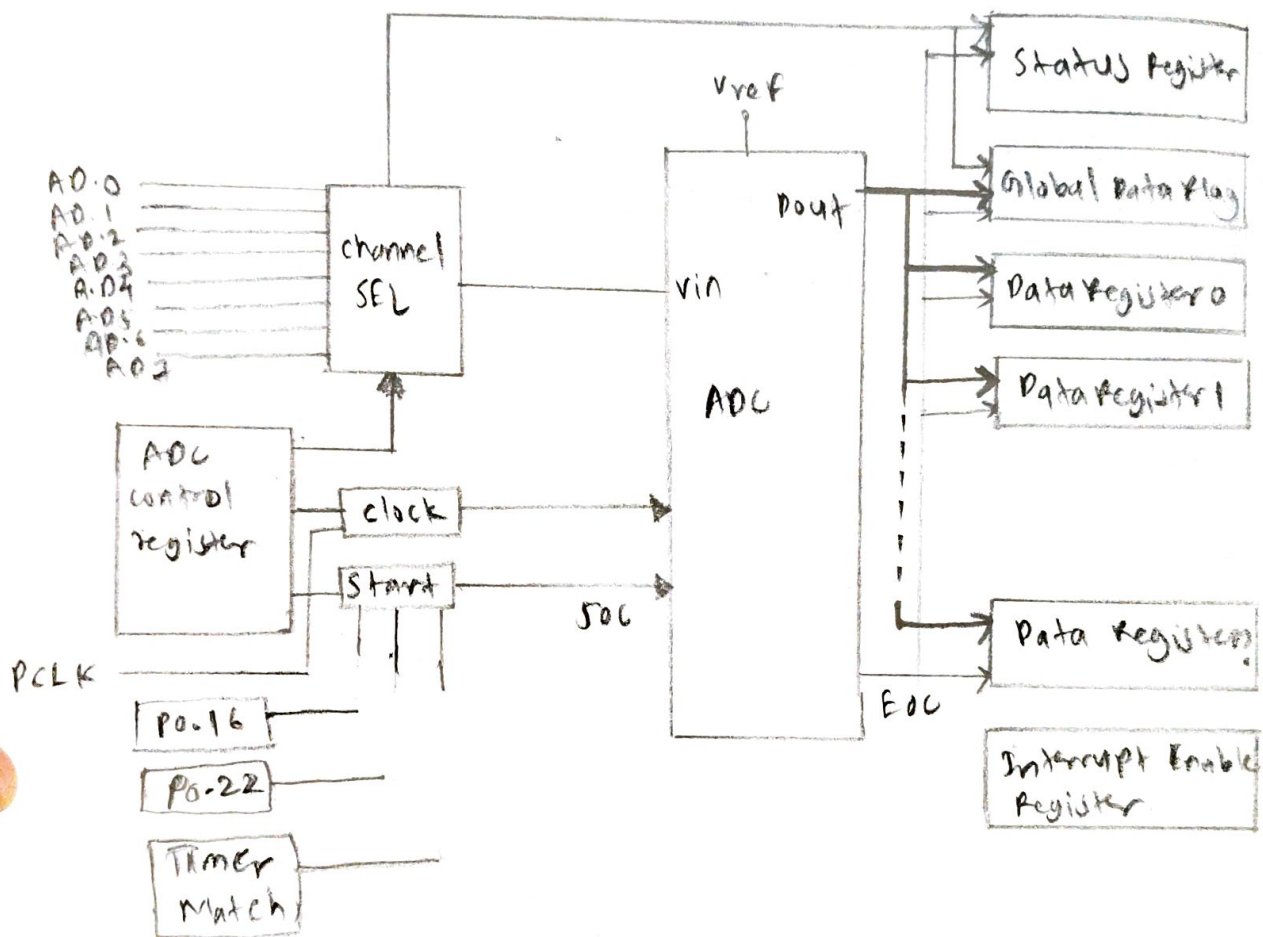


Fig. Interfacing of ADC with ARM7, LPC2148

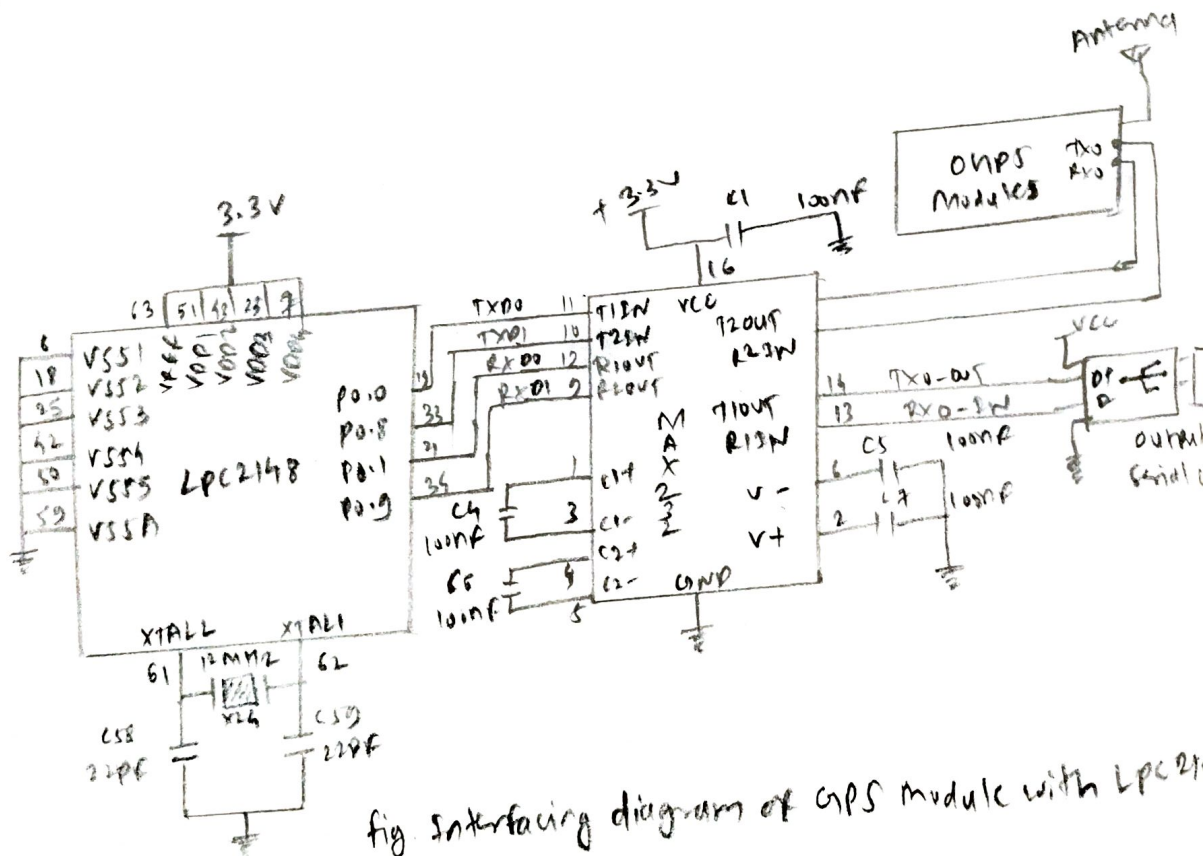




fig. Interfacing diagram of GPS module with LPC 2148.

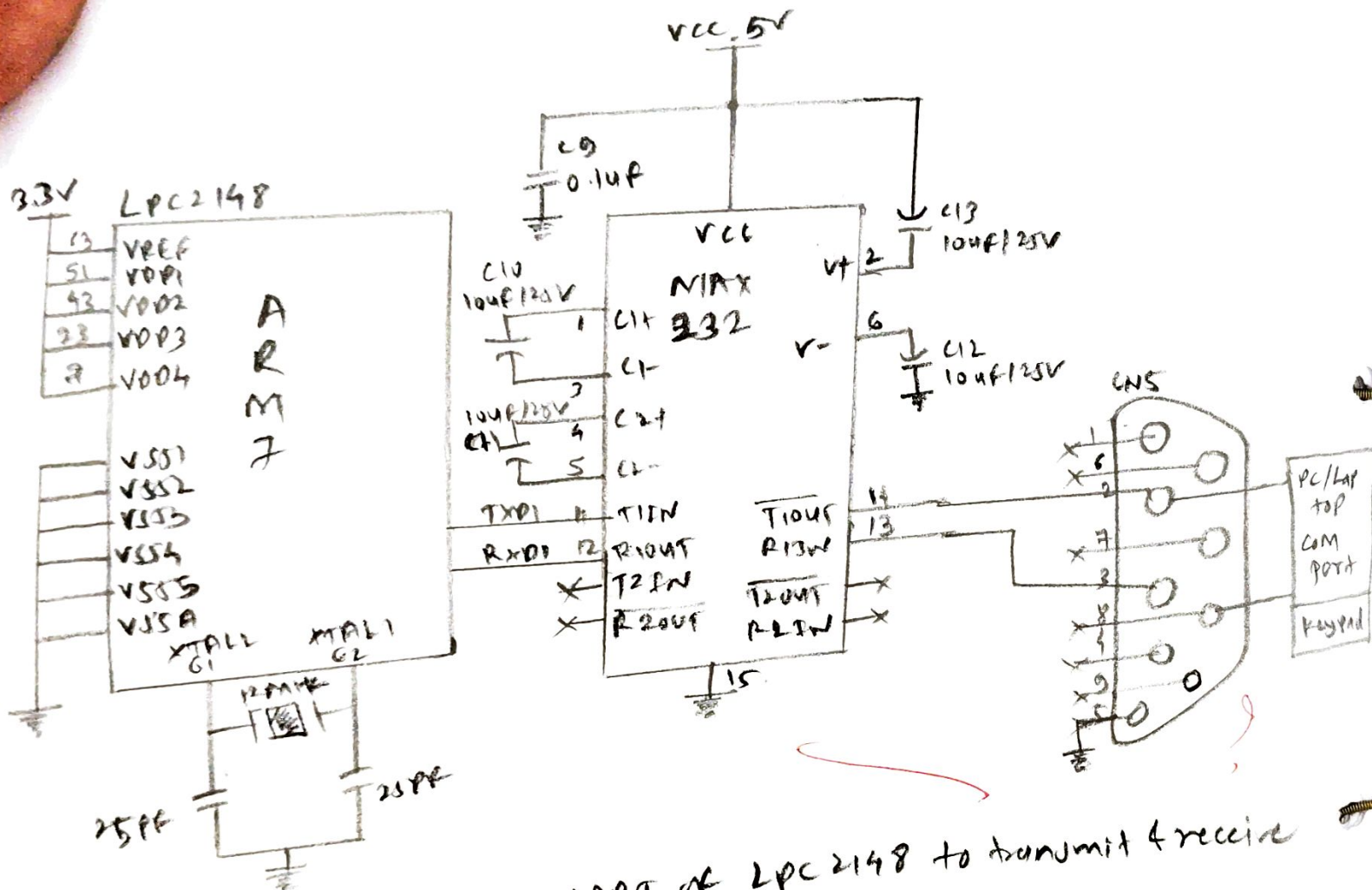


Fig. Interfacing UART of LPC2148 to transmit & receive message

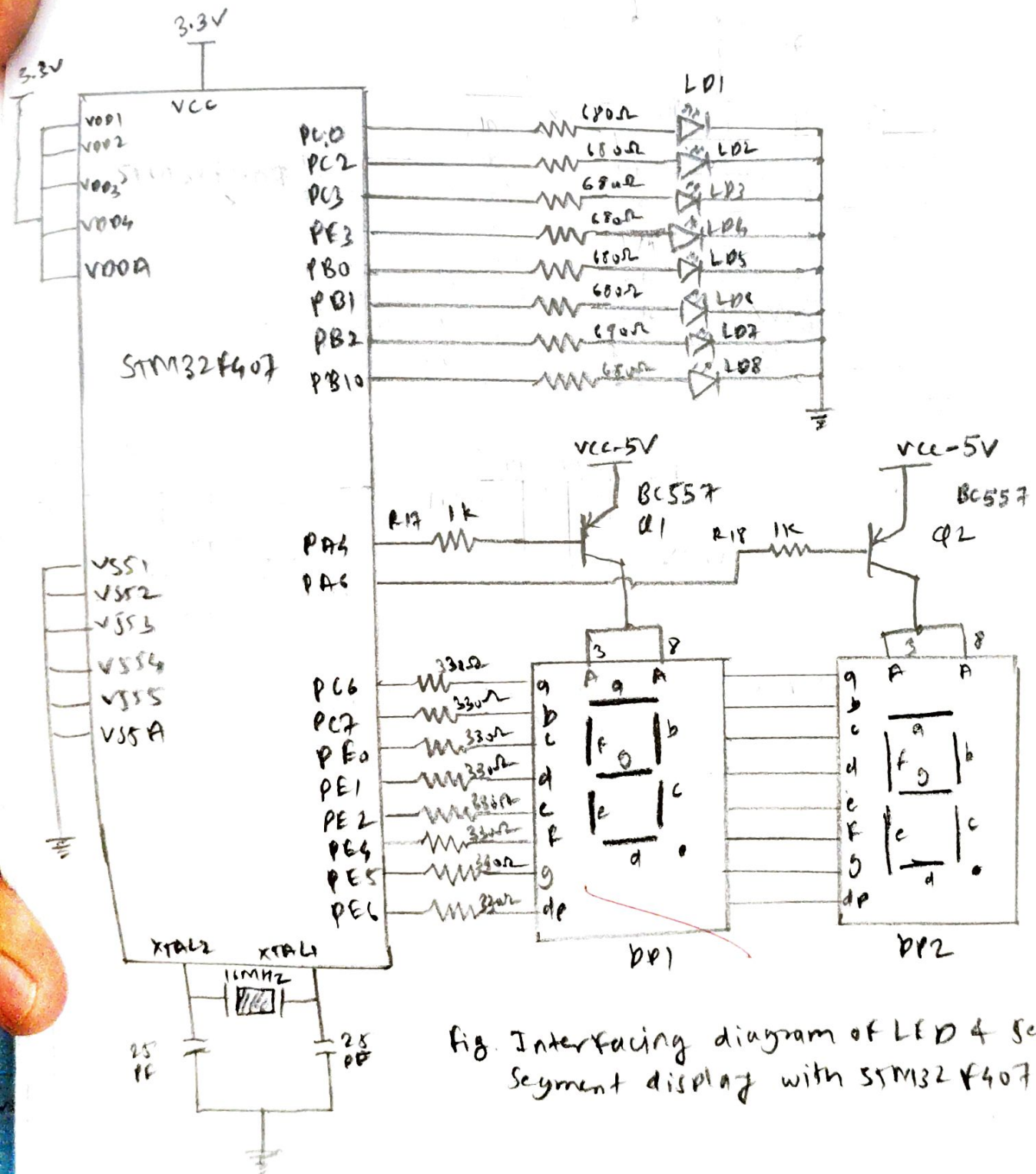


Fig. Interfacing diagram of LED & seven Segment display with STM32F407xx.

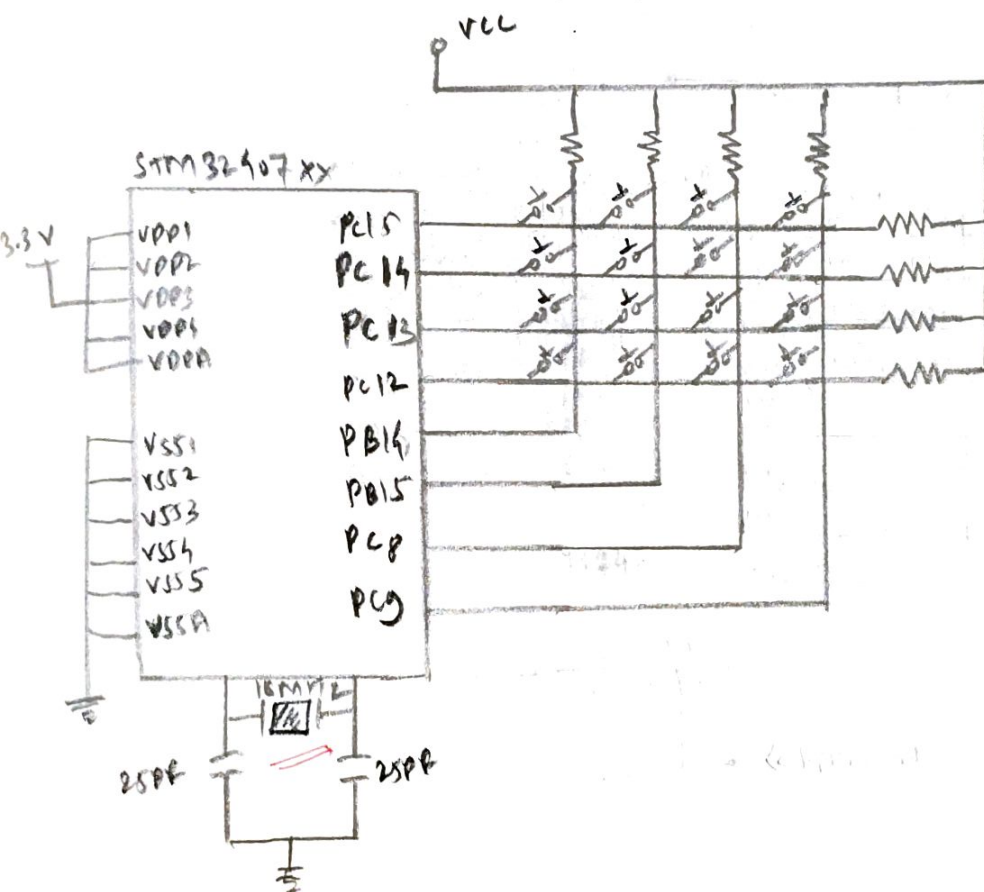


Fig. Keypad interfacing with ~~stm32f407xx~~ to transmit character using UART.

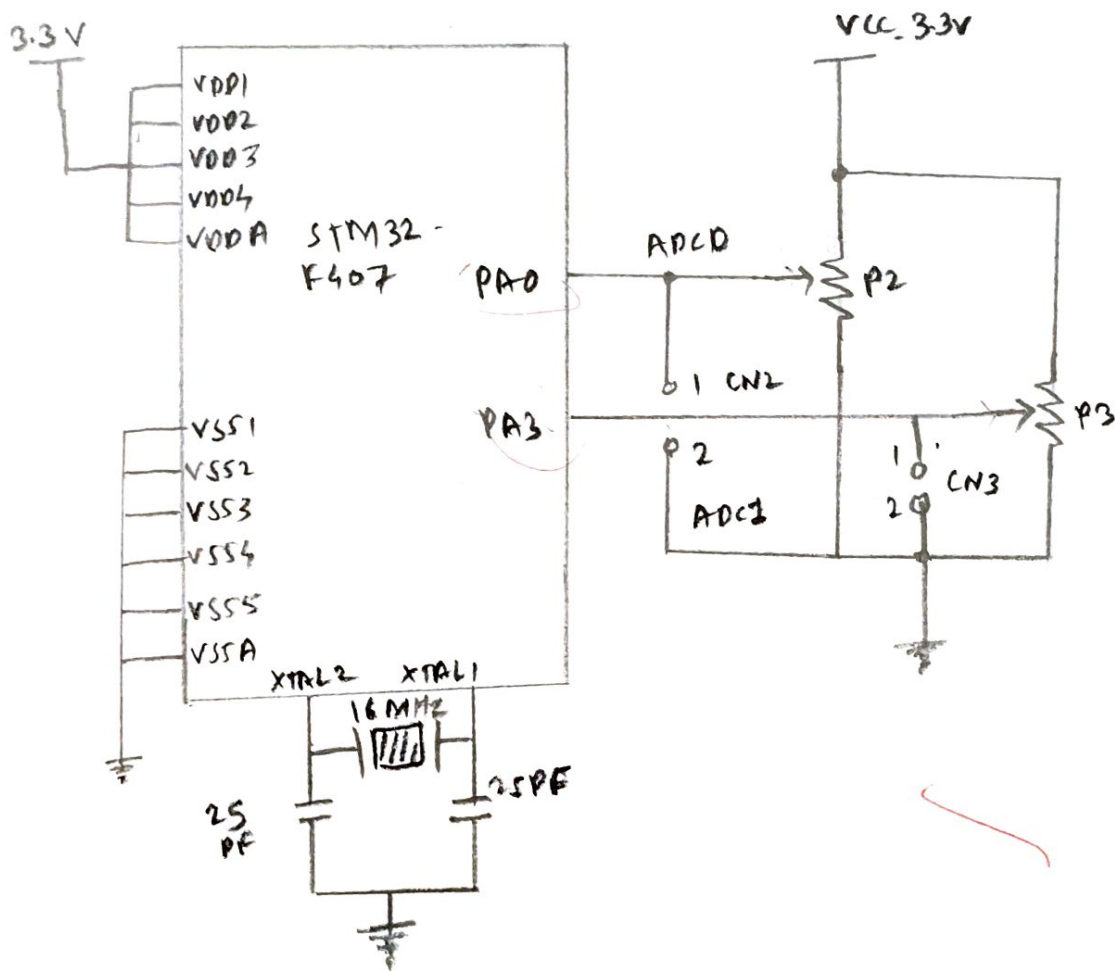


Fig. Interfacing diagram for ADC with STM32F407 microcontroller.

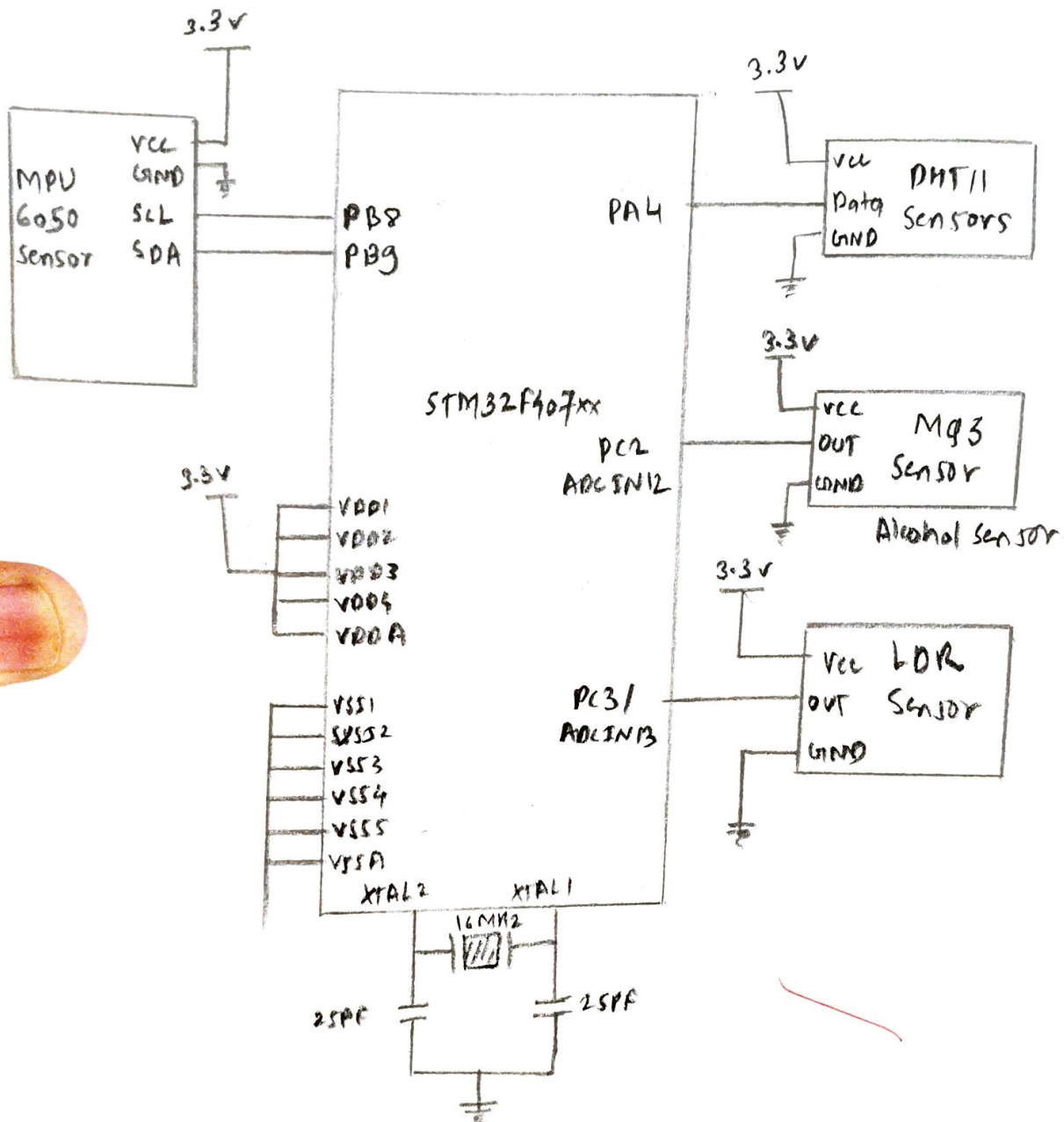


Fig. Interfacing of MPU6050 sensor & MQ3 alcohol sensor and DHT11 sensor and LDR sensor with STM32F407xx mc.

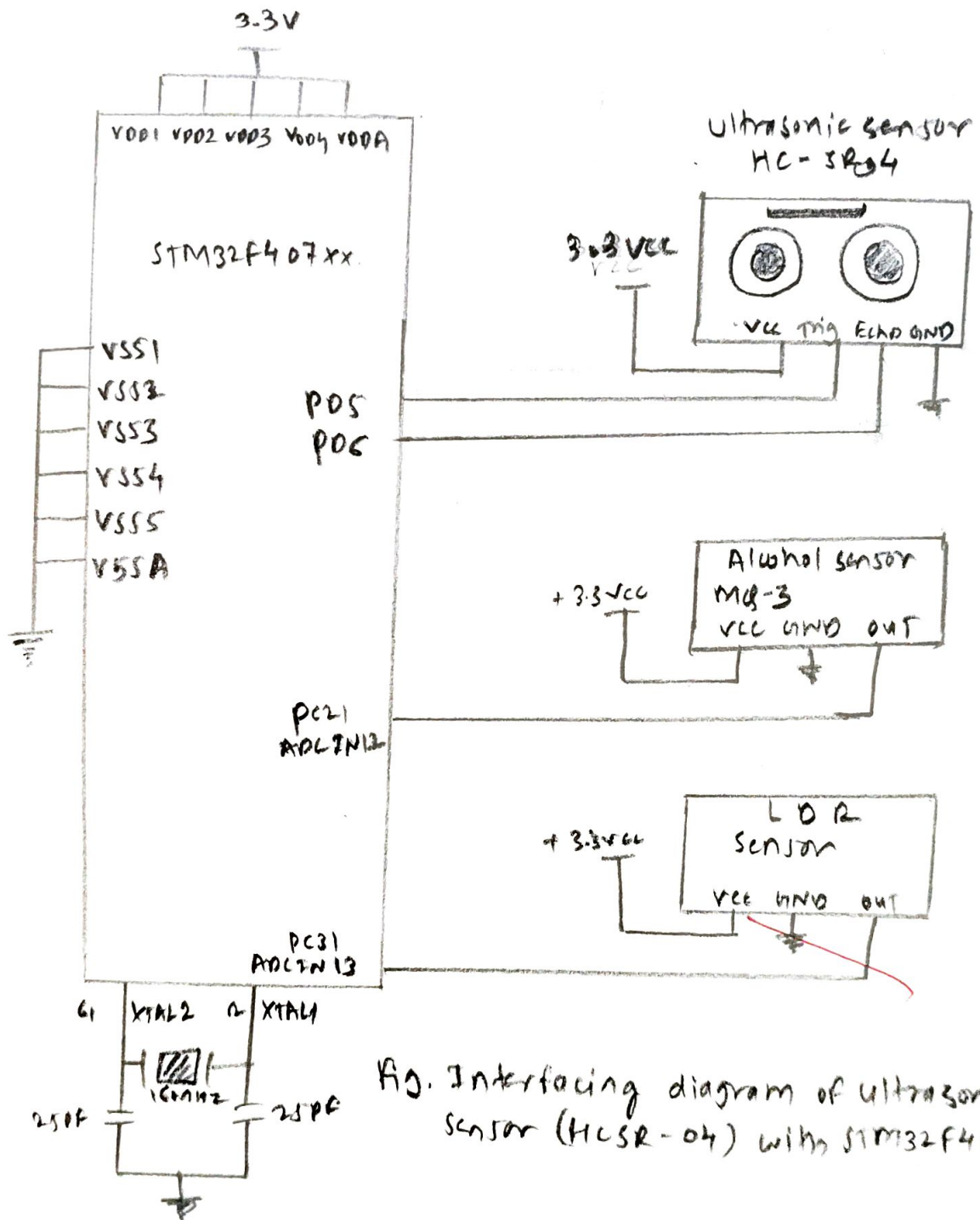


Fig. Interfacing diagram of ultrasonic sensor (HC-SR-04) with STM32F407.